Luke Trusheim

San Francisco Bay Area

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Education

California Polytechnic State University, San Luis Obispo — **Sep. 2025 – Jun. 2026**
Bachelor of Science in Computer Engineering | GPA: 3.92, Summa Cum Laude, 12x Dean's List — *San Luis Obispo, CA*

Experience

NZXT, Inc. — **Jul. 2025 – Sep. 2025**
Firmware Engineer Intern — *Remote*

- Designed, implemented, and validated a wireless lighting control system on Arduino Nano 33 BLEs
- Implemented multi-protocol firmware integrating BLE, UART, and SPI with a coprocessor handling 800 kHz timing
- Evaluated system architecture tradeoffs (BLE star vs. mesh topology, streaming vs. command-based animation)
- Authored report recommending architecture changes for NZXT to transition prototype toward manufacturable product

Computer Engineering Department — **Nov. 2024 – Present**
Computer Engineering Tutor — *San Luis Obispo, CA*

- Help students debug and understand microcontrollers, firmware, computer architecture, circuits, and digital design

Projects

Buddy Finder Compass | *ESP32, SPI, I2C* — **Jan. 2026 – Present**

- Developing handheld GPS compass devices on ESP32 that locate each other over point-to-point LoRa
- Architected multi-core FreeRTOS firmware with dedicated radio and navigation tasks communicating via queues
- Integrated SPI (LoRa), I2C (IMU), and UART (GNSS) peripherals with interrupt-driven async receive

Mesh-Connected Wildlife Camera Using Computer Vision | *Raspberry Pi, Python* — **Sep. 2025 – Mar. 2026**

- Rewrote open-source mesh network protocol in Python to work on ARM-based Linux systems (R. Pi Zero 2 W)
- ntegrated motion sensor, camera, solar panel and batteries with Raspberry Pi and computer vision models
- Wrote multithreaded control software with watchdog-based recovery handling GPIO, inference, and packet transfer

Lightweight Process Library | *C, x86-64 Linux* — **Jan. 2026**

- Implemented a user-level threading library in C with cooperative context switching and a round-robin scheduler
- Managed full thread lifecycle including creation, yielding, termination, and blocking with FIFO scheduling queues

Bare-Metal Digital Camera on STM32 | *STM32, SPI, I2C* — **May. 2025 – Jun. 2025**

- Developed bare-metal firmware for STM32-based digital camera system integrating camera, LCD screen, and SD card
- Implemented interrupt-driven DMA pipeline to capture and transfer real-time image data without frame loss
- Interfaced with peripherals using SPI and I2C and configured registers based on datasheet specifications

2025 WERC Environmental Engineering Competition | *Python, Arduino* — **Oct. 2024 – Apr. 2025**

- Collaborated with a multi-disciplinary team of 8 engineering students for the 2025 WERC Environmental Design Contest, winning first place by researching a DERMS solution that integrates solar power and hydrogen fuel cells
- Implemented a bench-scale by using Arduino microcontroller to monitor and control peripheral devices
- Designed Python algorithm to simulate solution on real data, demonstrating stable energy output and cost savings

Technical Skills

Languages: Python, C, C++, Bash, SystemVerilog
Developer Tools: VS Code, GitHub, Git, Make, Wireshark, Vivado, LTSpice
Technologies/Frameworks: Linux, FreeRTOS, ROS2, BLE, LoRa, GNSS, DMA
Electronics and Hardware: ESP32, STM32L4 (ARM), NRF52, SPI, I2C, UART, GPIO, PWM, oscilloscope, multimeter, logic analyzer, waveform generator, BJTs, MOSFETs, op-amps, datasheets, soldering